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Effect of Moisture Conservation Measures on Runoff, Soil Loss and Yield of Upland Rice

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ABSTRACT

Field experiments were conducted during 2002 and 2003 in light textured eroded sloping land to study the effect of moisture conservation measures on runoff, soil loss and yield of upland rainfed rice in Balijorenala watershed, Orissa. Total six treatments i.e. (i) T_1- fallow without any crop and conservation measures (control), (ii) T_2- crops grown without any conservation measures, (iii) T_3- vetiver grass planted in paired rows, (iv) T_4- miniature bund with vetiver grass, (v) T_5- miniature bund with natural grass and (vi) T_6- V-ditch with vetiver grass were taken for the study. The treatment T_6 was found to be superior to other treatments since it produced lowest average seasonal runoff and soil loss of 18.2% and 1.71 t/ha, respectively. On the other hand, the treatment T_1 was observed to generate highest amount of average seasonal runoff and soil loss with values of 28.0% and 6.13t/ha, respectively. The yield of rice was found to be maximum with treatment T_6 with a mean value of 19.1 q/ha whereas that for treatment T_2, it was the lowest (13.1 q/ha). Thus, there was 45.8% increase in rice yield due to the maximum utilization of stored soil moisture in treatment T_6 compared to treatment T_2. Considering the effectiveness in reducing runoff, soil loss and enhancing the crop yield, treatment T_6 is recommended as the best *in-situ* moisture conservation measures for rainfed upland farming system in sloppy land of Orissa.

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The state Orissa is bestowed with ample rainfall resources with average rainfall of 1450 m, 80% of which occurs concentrated during the monsoon season between June to September. During this period, about 50% of the annual rainfall comes from a few intense storms (Pisharoty, 1990). Water received from such intense storms is mostly lost as runoff, which also causes soil erosion. To reduce the waste of water, these could be several ways of enhancing rainwater use including *in-situ* conservation of rainwater in soil profile, conservation of rainwater in farm ponds and artificial groundwater recharge (Greb *et al.*, 1967; Narayana, 1979; Subbaiah, 1991). Generally storage of rainwater in soil profile is cheaper than any other methods of water conservation. In view of the dwindling water supplies and declining water table in many areas in Orissa, greater emphasis is now being given to conserve the *in situ* rainfall. In sloping watershed, conservation of rainwater in the soil profile is very crucial to reduce soil erosion. Runoff and soil loss are generally found to be more in fallow land without any moisture conservation measures.

Pujhari (1994) reported that in a fallow land, the runoff was reduced from 23.5 to 16.5% of rainfall and soil loss was reduced from 7.866 to 2.222 t/ha when the treatment of inter-terrace vetiver bunding was imposed. Earlier, several investigators have recommended the treatment of vetiver barriers as an effective soil and water conservation measure to control runoff and soil loss in a watershed. The aforesaid treatment is reported to reduce the runoff varying from 22 to 43% and soil loss varying from 20 to 60% over the farmers practice without taking any soil and water conservation measures (Subudhi *et al.*, 1998; Subudhi and Senapati, 1996; Channapa, 1994; Ranade *et al.*, 1997). Several researchers have also reported the mechanical measures like bunding, terracing, V-ditch etc. that can effectively reduce the runoff and soil erosion and thereby increase the crop yield. Bunding is found to increase the wheat and maize yield by 21.4 and 19.7%, respectively over the farmer's practice of broadcasting method without bunding system (Hadda and Sur, 1987). Radder *et al.* (1991) have similarly reported the increased yield of safflower and sorghum due to bunding in the sloping land of medium deep black soil of Karnataka. In the present study, an attempt was made to study the effect of mechanical as well as vegetative measures on runoff and soil loss and consequently to study their effects on the crop yield of a moderately sloping watershed of Orissa.

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MATERIALS AND METHODS

The present study was carried on in Balijorenala watershed of West Central Table Land Zone of Orissa during *kharif* 2002 and 2003. The area comes under the sub humid region of the district of Sambalpur, Orissa and is situated at 20° 232 N latitude, 83°392 E longitude and at an altitude of 127.35m AMSL. The topography of the watershed is undulating and inter-spersed with hills and valleys. Locally the land is classified as Att (unbunded upland). Soil of the experimental plots in the watershed is sandy loam with sand, silt and clay content of 66.5, 18.5 and 15%, respectively. Average value of pH and bulk density of the soil are 6.3 is 1.65 g/cm^3, respectively. The average annual rainfall and pan evaporation of the study area are 1425 and 1512 mm, respectively. The mean minimum and maximum temperatures are 12.2°C in December and 42°C in May. The mean monthly relative humidity ranges from 45.5 to 92.5%.

The study was undertaken with six treatments i.e. (i) T_1- fallow without any crop and moisture conservation measures (control), (ii) T_2- crops grown in the plots but without any moisture conservation measures, (iii) T_3- vetiver grass (*Vetiveria zizinoides*) planted in paired rows, (iv) T_4- miniature bund with vetiver grass planted at the upstream end of the bund, (v) T_5- miniature bund with natural grass planted at the upstream end of the bund and (vi) T_6- V-ditch with vertiver grass planted in the ditch. Size of each plot was 32 x 12 m, having 3% slope. In treatment T_3, The vetiver grass was planted in three strips of 30 cm width. The treatments T_3 to T_6 were imposed in the field at horizontal interval of 8m. Early variety of rice (Pathara) was grown on contour in the experimental plots. Rice was sown on 2nd and 4th July during 2002 and 2003, respectively. It was harvested on 2nd and 4th October during 2002 and 2003, respectively. Recommended dose of fertilizers were applied to the crop for all treatments. Plant protection measures were taken as and when required. During the cropping season, daily runoff was monitored with the help of multislot divisors. Water samples from the runoff collection tanks were collected periodically to analyse the sediment content (Pujhari, 1994). The data on daily rainfall during the crop growing season was collected from the nearest meteorological observatory, which is only 1 km from the study site set up by the soil conservation department. Soil samples were collected from 15, 30 and 45 cm depth of each plot once in a week and the moisture content was determined gravimetrically. The yield of rice was measured for each plot at the harvest time.

RESULTS AND DISCUSSION

Runoff Computation

There was 847 mm rainfall during the rice-growing season of 2nd July to 2nd October during 2002. During this period, 226.2, 207.5, 191.4, 167, 178 and 157 mm runoff were generated from the rice field with treatments T_1 to T_6 in sequences. Similarly, the rice-growing period of 2003 (July 4th to October 4th), received 757 mm rainfall. This resulted in generating 221.0, 182.0, 161.2, 142.4, 150.0 and 135.0 mm runoff by treatments T_1 to T_6 in sequences (Table 1). Treatment T_6 having V-ditch with vetiver grass planting in it was found to produce lowest runoff of 157.0 and 135.0 mm during 2002 and 2003, respectively that is 18.6 and 17.8% of the seasonal rainfall in these two years. Next best treatment was T_4 that was found to produce 167.0 and 142.4 mm runoff during 2002 and 2003, respectively (19.7 and 18.8% of seasonal rainfall during these two years). However, treatment T_1 produced highest amount of runoff contributing 26.7 and 29.2% of total seasonal rainfall during 2002 and 2003, respectively (Table 2).

Several investigators have reported that vetiver barrier is effective in minimizing the surface runoff and soil loss to the lowest amount compared to all other vegetative barriers. Subudhi and Senapati (1996) observed a reduction of runoff of 22, 19 and 18% in rice fields

Table 1. Treatment details as followed in the experiment study

S.No.	Treatment	Treatment details
1	T1	Fallow without any crop and moisture conservation measure (control)
2	T2	Crops grown in the plots but without any conservation measure
3	T3	Vetiver grass (Vetiveria zizinoides) planted in paired rows
4	T4	Miniature bund with vetiver grass planted at the upstream end of the bund
5	T5	Miniature bund with natural grass planted at the upstream end of the bund
6	T6	V-ditch with vetiver grass planted in the ditch

Table 2. Effect of different conservation measures on seasonal runoff, soil loss and yield of rice

Treatment	Runoff, % of rainfall			Soil loss, t/ha			Grain yield, q/ha		
	2002	2003	Average	2002	2003	Average	2002	2003	Average
T1	26.7	29.2	28.0	6.04	6.21	6.12	---	----	---
T2	24.5	24.0	24.3	3.34	3.30	3.32	12.6	13.6	13.1
T3	22.6	21.3	22.0	2.42	2.32	2.37	14.6	15.4	15.0
T4	19.7	18.8	19.3	1.87	1.79	1.83	16.8	18.2	17.5
T5	21.0	19.8	20.4	1.96	1.91	1.93	15.7	16.7	16.2
T6	18.6	17.8	18.2	1.75	1.66	1.71	18.4	19.8	19.1
CD (P=0.05)	7.1	10.2	9.3	2.1	1.9	1.95	1.3	1.5	1.45

In 2002 and 2003, the seasonal runoff and soil loss were measured from 2nd July to 2nd October and 4th July to 4th October, respectively.

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when planted with barriers of vetiver, napier and sabai grass, respectively. In another experiment, Subudhi *et al.* (1998) found out a reduction of 35% runoff over the farmer's practice without taking any conservation measures. The study further indicated that the runoff amount decreased for all the treatments (except T_1) in 2003 compared to 2002 since there was a good establishment of grasses on the miniature bund and V-ditches in 2003 (2nd year of the establishment) than 2002 (1st year of establishment). On the other hand, the fallow field without conservation measure (T_1) had a deteriorating condition progressively since the runoff was found to increase in 2003 than 2002 (Table 2). Variation of weekly runoff, weekly soil loss and tests of significance in terms of critical difference (CD at P = 0.05) obtained from computation of both years of weekly runoff and soil loss are presented in Table 3. It is found from this table that treatment T6 was the best to reduce the runoff and soil loss compared to all other treatments.

Table 3. Weekly variation* of runoff and soil loss and their statistical tests of significance as influenced by various treatments

Treatment	Variation of runoff, mm	Variation of soil loss, t/ha
T1	3.4-42.0	0.054-1.122
T2	3.0-35.8	0.085-0.621
T3	2.7-33.9	0.043-0.533
T4	2.0-30.1	0.031-0.362
T5	2.5-31.9	0.039-0.383
T6	1.9-27.9	0.028-0.331
CD (P=0.05)	0.61-7.52	0.013-0.064

*Weekly variation represents the average of the standard meteorological weeks from 27th to 39th of two years data of 2002 and 2003.

Soil Loss

Treatment T_1 recorded highest amount of soil loss (6.04 and 6.21 t/ha) during 2002 and 2003, respectively, followed by Treatment T_2. On the other hand treatment T_6 significantly produced the lowest amount of soil loss (1.75 and 1.66 t/ha) during 2002 and 2003, respectively. Thus, on an average there was 72.1% decrease in soil loss by treatment T_6 over the treatment T_1. Subudhi *et al.* (1998) reported 60.0% decline of soil loss over the broadcasted field without any conservation measures. Channapa (1994) reported 43.0% decrease of soil loss due to vetiver planting over the farmer's practice with no measure. Seasonal values of soil loss by various treatments in both 2002 and 2003 are presented in Table 2. Weekly variation of soil loss (Table 3) was found to be the lowest for Treatment T_6 (0.028 to 0.331 t/ha) that is observed to be significantly the lowest amongst all other treatments. Like runoff, the values of soil loss were observed to be lower for all the treatments (except T_1) for the year 2003 than 2002 revealing the fact the good establishment of the grasses in the successive years had a causative factor on the reduction of the soil loss.

A linear model was observed between the rainfall and runoff with values of coefficient of determination (R^2) of more than 0.84 for all the treatments. Soil loss was also found to be dependent on rainfall as well as runoff irrespective of treatments imposed. Linear relationship (Eq. 1) between soil loss vs. runoff and soil loss vs. rainfall were observed for all the treatments with values of R^2 of more than 0.74.

$$y = A x + B \quad \ldots(1)$$

In case of rainfall-runoff process, the parameters y and x of Eqn. (1) are runoff and rainfall, respectively and are given in mm. In case of rainfall-soil loss process, the

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parameters y and x of Eq. 1 are soil loss and rainfall, respectively and are expressed in units of t/ha and mm, respectively. Similarly for runoff-soil loss process, the parameters y and x of Eqn. (1) are soil loss and runoff, respectively and are expressed in units of t/ha and mm, respectively. The values the coefficients A and B of the linear model (Eq. 1) for various processes are given in Table 4, which helps in the determination of runoff and soil loss.

Soil Moisture Content

Table 5 represents the average values (average of years and over entire crop growing period) of moisture content (per cent dry weight basis) in the soil profile in 0-15, 15-30 and 30-45 cm root zone depth of the crop. Data of this table indicate that soil moisture content is maximum for all the profile layers for treatment T_6. The mean value of soil moisture content for treatment T_6 was observed as 15.0% whereas that for treatment T_1 was 12.7%. The reason for obtaining higher moisture contents in treatment T_6 might be due to higher infiltration of runoff since more runoff was retained by the V-ditch with vetiver planting than any other treatments. On the other hand the fallow land without any conservation measure (treatment T_1) produced more runoff as it had no barrier to retain the runoff flowing over the sloppy land and so the soil moisture content measured for it was lowest for all the profile layers.

Grain Yield

Among all the treatments, T_6 was found to produce significantly highest rice yield of 18.4 and 19.8 q/ha during 2002 and 2003, respectively. This followed the treatment T_4 with average yield of 17.5 q/ha. However, treatment T_2 with no conservation measure yielded lowest rice yields of 12.6 and 13.6 q/ha with a mean value of 13.1 q/ha (Table 2). The mean rice yield of treatment T_6 was recorded to be 45.8% higher over that of treatment T_2. The reason for obtaining highest yield by treatment T_6 over other treatments may be due to lowest runoff and soil loss and consequently due to higher accumulation of soil moisture in the root zone layers during the entire crop growth period. Significant increase of rice yield has been reported by several authors due to vetiver barriers over the farmer's practice of broadcasting without any conservation measures. Compared to farmer's practice, vetiver barrier treatment was reported to yield 72.6% more rice yield by Pujhari (1994) and 93% more rice yield by Subudhi *et al.* (1998).

Table 4. Coefficients of linear models* of the parameters of rainfall, runoff and soil loss

Treatment	Rainfall-runoff process			Rainfall-soil loss process			Runoff-soil loss process		
	A	B	R^2	A	B	R^2	A	B	R^2
T1	0.263	0.252	0.85	0.007	0.006	0.78	0.027	-0.001	0.80
T2	0.226	0.268	0.84	0.004	0.003	0.80	0.017	-0.001	0.77
T3	0.210	0.115	0.86	0.003	-0.014	0.77	0.015	-0.016	0.75
T4	0.190	-0.324	0.85	0.002	-0.004	0.78	0.012	-0.005	0.78
T5	0.198	-0.067	0.85	0.002	-0.001	0.75	0.012	-0.001	0.82
T6	0.178	-0.189	0.87	0.002	-0.003	0.77	0.011	-0.002	0.81

*The form of the linear model is y = Ax + B, where y = runoff (mm) and x = rainoff (mm) in rainfall-runoff process; y = soil loss (t/ha) and x = rainfall (mm) in rainfall-soil loss process; y = soil loss (t/ha) and x = runoff (mm) in runoff-soil loss process. Values in parenthesis represents coefficient of determination ((R^2)).

Table 5. Soil moisture content in various profile layers by different treatments

Treatment	Soil moisture content, %*			
	0-15 cm	15-30 cm	30-45 cm	Mean
T1	11.2	12.7	14.2	12.7
T2	12.2	13.5	14.8	13.5
T3	12.8	13.4	15.1	13.8
T4	13.5	15.2	15.6	14.8
T5	13.1	14.8	15.2	14.4
T6	13.8	15.2	15.9	15.0

*Values represent the average over 2002 and 2003 and over the entire crop-growing period

CONCLUSIONS

The study revealed that fallow land as well the land covered with crops but without any conservation measure produce more runoff and soil loss compared to land covered with crop residue along with conservation measures. In the experiment, V-ditch with vetiver planting (treatment T_6) was observed to produce lowest runoff (mean value of 18.2% of seasonal rainfall) and soil loss (1.71 t/ha) compared to the other treatments. Runoff in the rice field was reduced by 35% with the above-mentioned treatment over the fallow land without conservation measure (treatment T_1). Thus, on an average there was 72.1% decrease in soil loss by treatment T_6 over the treatment T_1. Further, treatment T_6 resulted in an average increase of 45.8% rice grain yield over treatment T_2 without any conservation measure. Considering the effectiveness in controlling runoff, soil loss and increasing the crop yield, V-ditch with vetiver planting is recommended as the best treatment for light textured, eroded sloping watershed of Orissa.

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